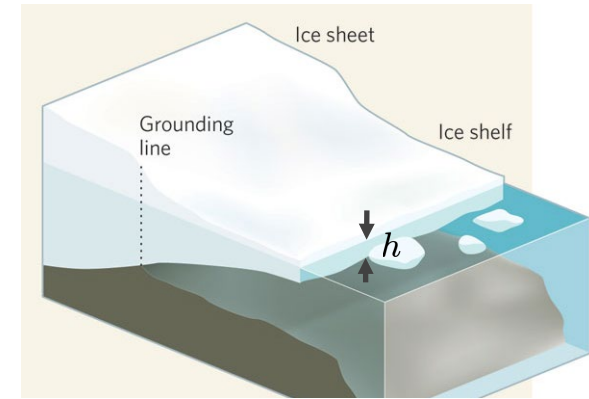
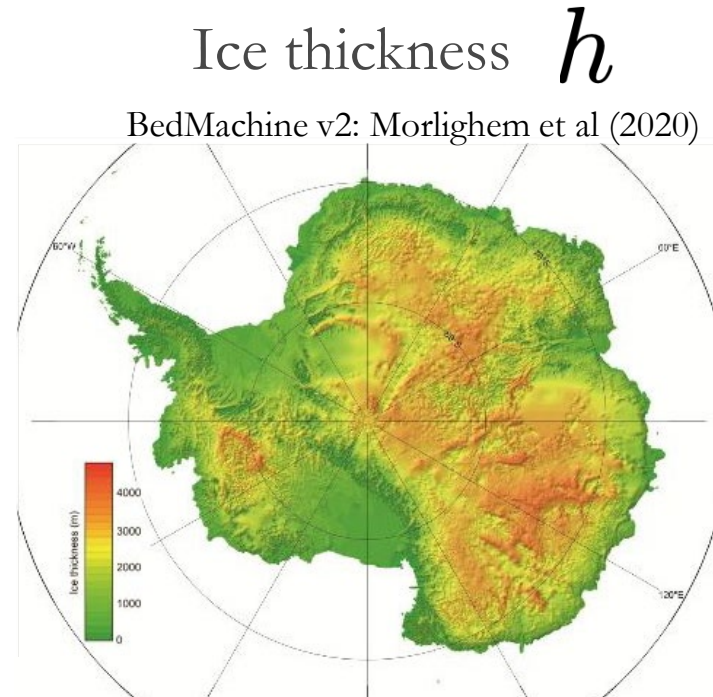
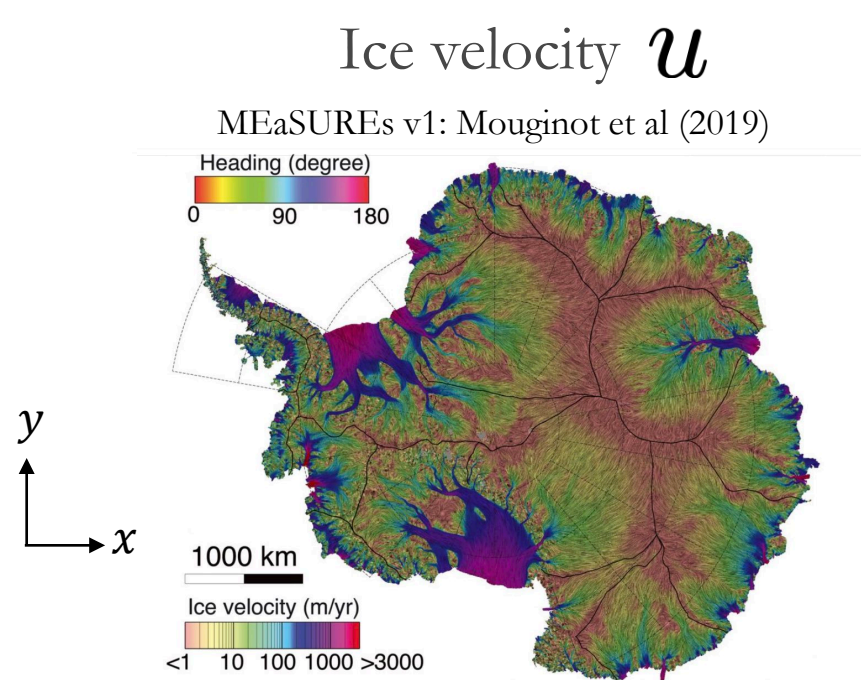


PINN Project- GlaMacLes

Ice shelves: inversion of **viscosity**

Data:



Equation:

Ice effective viscosity: impossible to measure

$$\underbrace{\frac{\partial}{\partial x} \left(4\mu h \frac{\partial u}{\partial x} + 2\mu h \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial y} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right)}_{\text{viscous stress}} = \underbrace{\rho_i g \left(1 - \frac{\rho_i}{\rho_w} \right) h \frac{\partial h}{\partial x}}_{\text{gravity}}$$

$$\underbrace{\frac{\partial}{\partial y} \left(4\mu h \frac{\partial v}{\partial y} + 2\mu h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right)}_{\text{viscous stress}} = \underbrace{\rho_i g \left(1 - \frac{\rho_i}{\rho_w} \right) h \frac{\partial h}{\partial y}}_{\text{gravity}}$$

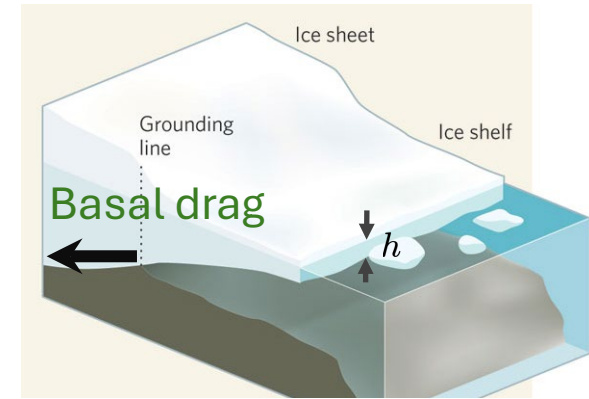
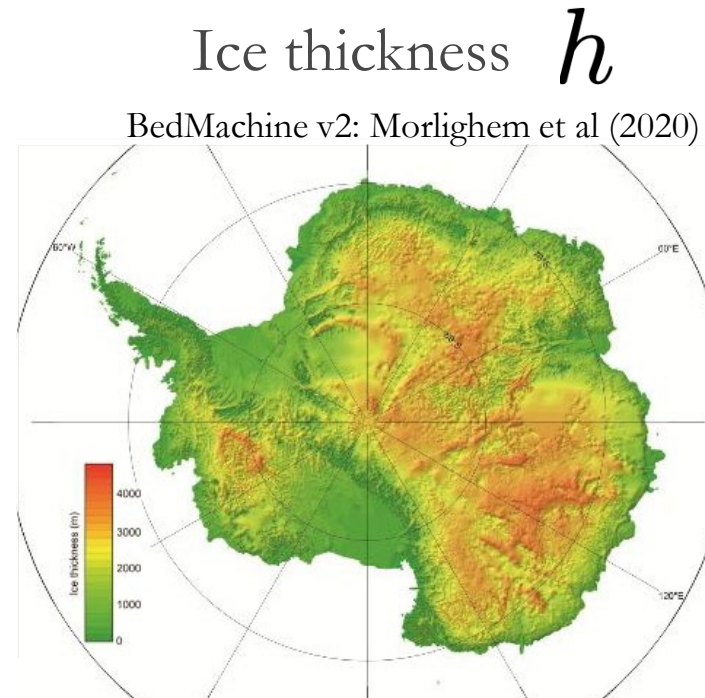
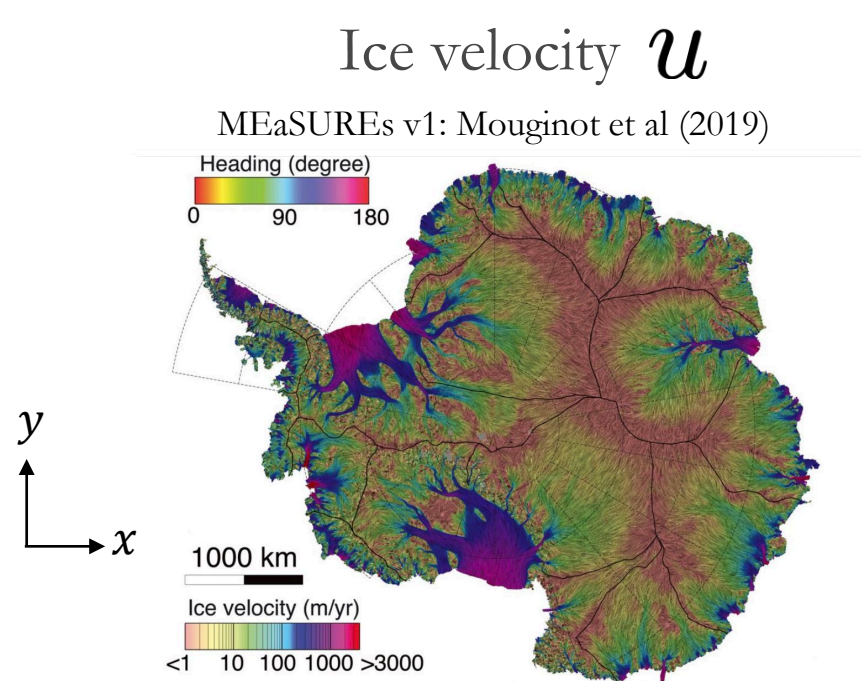
Morland (1987), MacAyeal (1989)

u, v : horizontal velocity
 ρ_i : ice density
 ρ_w : sea water density
 h : ice thickness

measurable

Ice sheet: inversion of both **viscosity** and **basal traction**

Data:



Equation:

Ice effective viscosity: impossible to measure

Morland (1987), MacAyeal (1989)

$$\frac{\partial}{\partial x} \left(4\mu h \frac{\partial u}{\partial x} + 2\mu h \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - \tau_{bx} = \rho_i g h \frac{\partial s}{\partial x}$$

$$\frac{\partial}{\partial y} \left(4\mu h \frac{\partial v}{\partial y} + 2\mu h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - \tau_{by} = \rho_i g h \frac{\partial s}{\partial y}$$

Basal traction: impossible to measure

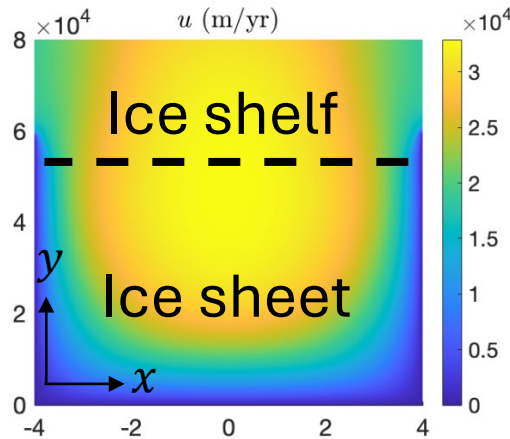
u, v : horizontal velocity
 ρ_i : ice density
 h : ice thickness
 s : surface elevation

measurable

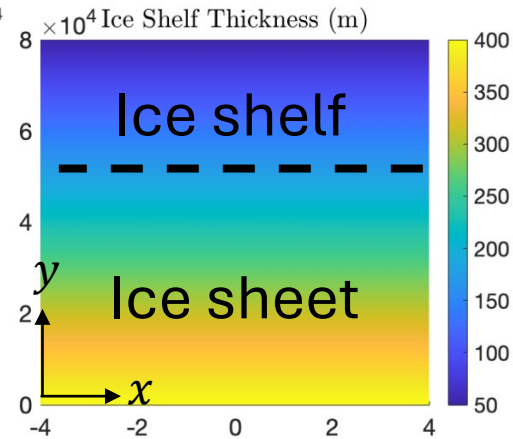
Ice sheet: inversion of both **viscosity** and **basal traction**

Data:

Ice velocity u

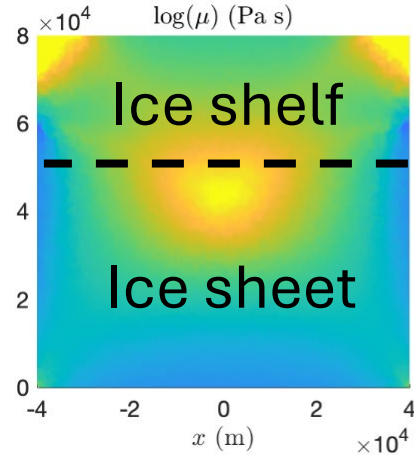


Ice thickness h

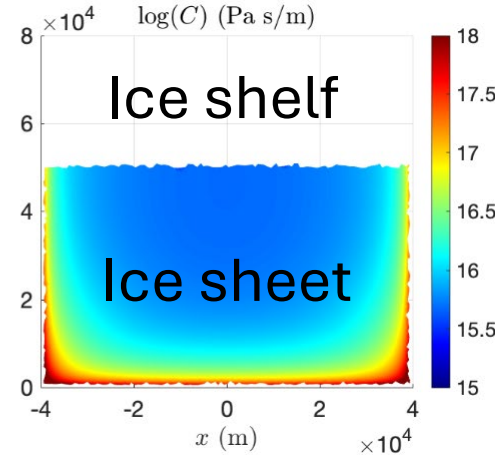


Predict (infer):

Ice viscosity μ



Basal traction C



Equation:

Ice effective viscosity: impossible to measure

$$\begin{aligned} \frac{\partial}{\partial x} \left(4\mu h \frac{\partial u}{\partial x} + 2\mu h \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cu &= \rho_i g h \frac{\partial s}{\partial x} \\ \frac{\partial}{\partial y} \left(4\mu h \frac{\partial v}{\partial y} + 2\mu h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cv &= \rho_i g h \frac{\partial s}{\partial y} \end{aligned}$$

Basal traction: impossible to measure

Morland (1987), MacAyeal (1989)

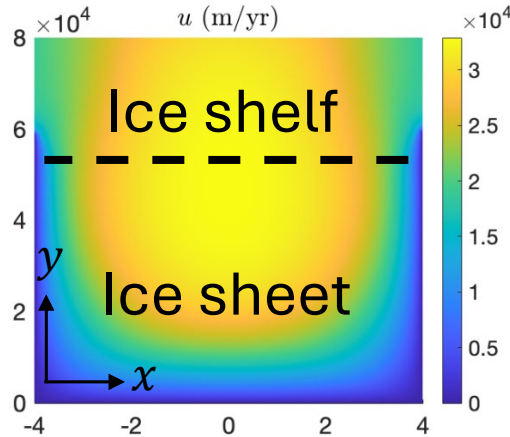
u, v : horizontal velocity
 ρ_i : ice density
 h : ice thickness
 s : surface elevation

measurable

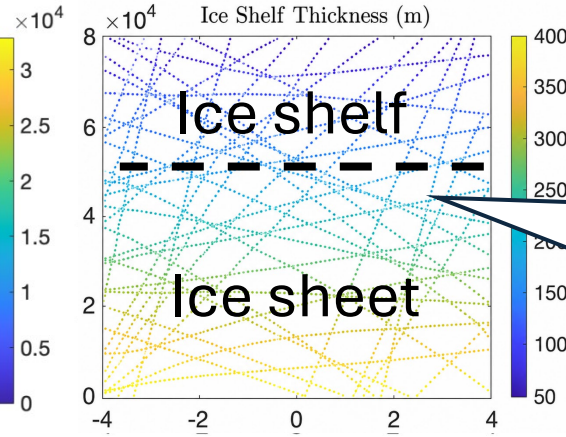
But thickness measurement is actually sparse & noisy!

Data:

Ice velocity u

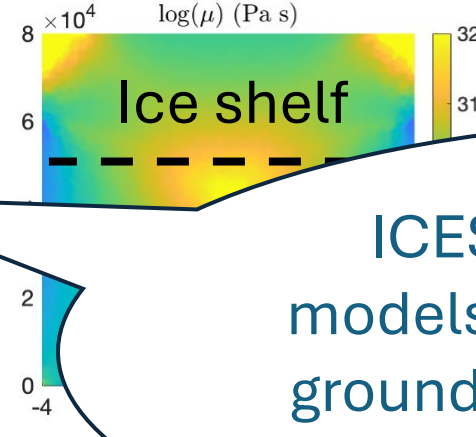


Ice thickness h

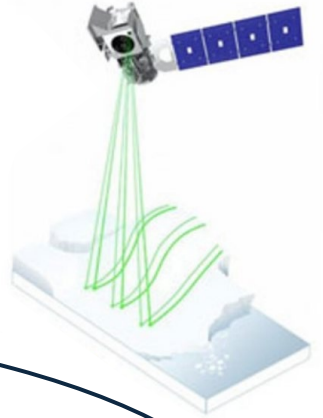
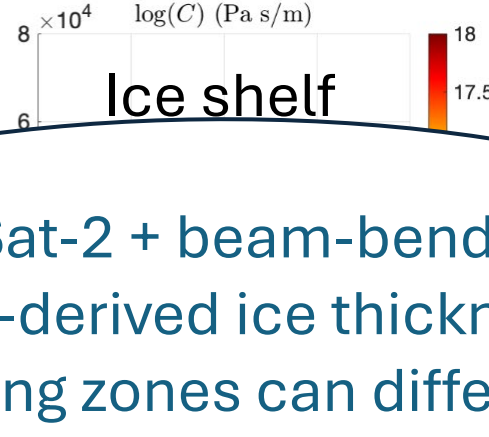


Predict (infer):

Ice viscosity μ



Basal traction C



ICESat-2 + beam-bending models-derived ice thickness in grounding zones can differ from BedMachine by up to 10%!

Equation:

Ice effective viscosity: impossible to measure

$$\begin{aligned} \frac{\partial}{\partial x} \left(4\mu h \frac{\partial u}{\partial x} + 2\mu h \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cu &= \rho_i g h \frac{\partial s}{\partial x} \\ \frac{\partial}{\partial y} \left(4\mu h \frac{\partial v}{\partial y} + 2\mu h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cv &= \rho_i g h \frac{\partial s}{\partial y} \end{aligned}$$

Basal traction: impossible to measure

u, v : horizontal velocity
 ρ_i : ice density
 h : ice thickness
 s : surface elevation

measurable

Ice sheet: inversion of both **viscosity** and **basal traction** and **thickness**!

Data:

Predict (infer):

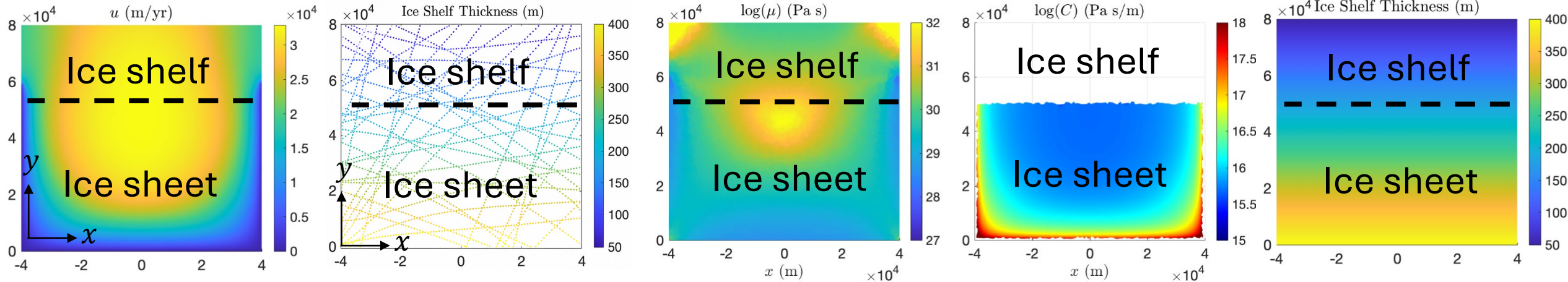
Ice velocity u

Ice thickness h

Ice viscosity μ

Basal traction C

Ice thickness h



Equation:

Ice effective viscosity: impossible to measure

Morland (1987), MacAyeal (1989)

$$\begin{aligned} \frac{\partial}{\partial x} \left(4\mu h \frac{\partial u}{\partial x} + 2\mu h \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cu &= \rho_i g h \frac{\partial s}{\partial x} \\ \frac{\partial}{\partial y} \left(4\mu h \frac{\partial v}{\partial y} + 2\mu h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(\mu h \frac{\partial u}{\partial y} + \mu h \frac{\partial v}{\partial x} \right) - Cv &= \rho_i g h \frac{\partial s}{\partial y} \end{aligned}$$

Basal traction: impossible to measure

u, v : horizontal velocity
 ρ_i : ice density
 h : ice thickness
 s : surface elevation

measurable

DIFFICE.jax v1.0:

DIFFerentiable neural-network solver for data assimilation of ICE shelves written in JAX.

 DIFFICE_jax

INVERSE PROBLEMS:
Isotropic viscosity inversion
Anisotropic viscosity inversion

EXTENDED-PINNS:
Why we need XPINNs
Loss function for XPINNs

TUTORIAL:
Overview
Forward problem setup
File descriptions
Results

TRAINING DATA:
Data source
Data Format

REAL-DATA EXAMPLES:
Overview
Isotropic case via PINNs
Isotropic case via XPINNs

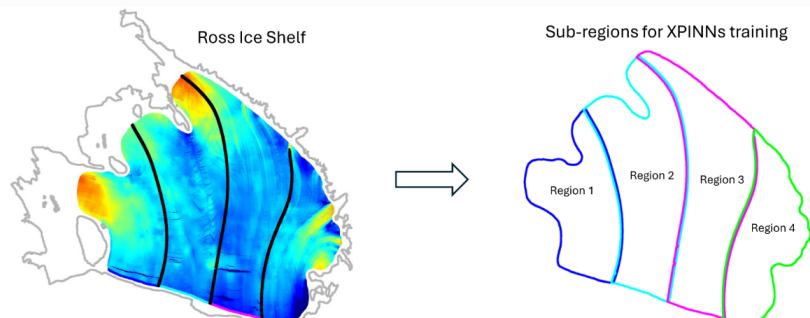
 / DIFFICE_jax documentation

 [Edit on GitHub](#)

DIFFICE_jax documentation

Introduction

`DIFFICE_jax` is a Python package that solves the depth-integrated Stokes equation for ice shelves, and can be adopted for ice sheets by modifying the partial differential equations (PDE) in the neural network loss function. It uses PDEs to interpolate discretized remote-sensing data into meshless and differentiable functions, and infer ice shelves' viscosity structure via back-propagation and automatic differentiation (AD). The algorithm is based on physics-informed neural networks (PINNs) and implemented in JAX. The `DIFFICE_jax` package involves several advanced features in addition to vanilla PINNs algorithms, including collocation points resampling, non-dimensionalization of the data and equations, extended-PINNs (XPINNs) (see figure below), viscosity exponential scaling function, which are essential for accurate inversion. The package is designed to be user-friendly and accessible for beginners. The Github repository also provides Colab Notebooks for both the synthetic and real ice-shelf examples in the `tutorial` and `examples` folders, respectively.



2D SSA:

https://github.com/YaoGroup/DIFFICE_jax

- Synthetic case
- Real data case

Package can be installed via

```
python -m pip install DIFFICE_jax
```



Wang, Lai, Prior, Cowen-Breen, Science (2025)
Wang & Lai, Journal of Open Source Software (2025)
Wang & Lai, Journal of Computational Physics (2024)